

A. Advising

time limit per test: 1 second memory limit per test: 256 megabytes

In this problem, there are N courses in the curriculum and M requirements of the form "Course A has to be completed before course B ".

Your task is to find an order in which you can complete the courses. If there are multiple valid order, you may print any of them. If no such valid order exists, then print -1 .

Input

The first line contains an integer T ($1 \leq T \leq 10^5$), denoting the number of test cases.

For each test case, the first line of its input contains two integers N and M ($1 \leq N, M \leq 10^5$), denoting the number of courses and requirements. This is followed by M lines, each containing two integers A and B ($1 \leq A, B \leq N$) denoting that course A has to be completed before course B .

It is guaranteed that the sum of N and M over all test cases will not exceed $2 \cdot 10^5$.

Output

For each test case, print an order in which you can complete the courses. If there is no valid order, then print -1 .

Examples

Input

```
1
6 6
4 2
5 6
1 4
2 3
6 3
1 6
```

Output

```
1 5 4 6 2 3
```

Input

```
3
1 1
1 1
6 5
4 3
1 3
3 6
6 4
2 3
4 2
3 1
4 1
```

Output

```
-1
-1
2 3 4 1
```

B. A Football Match

time limit per test: 2 seconds memory limit per test: 1024 megabytes

There is an intense football match going on between Robots and Humans. However, things aren't as simple as they seem — the Robots have disguised themselves to look exactly like Humans! From the outside, it's impossible to tell who is a Robot and who is a Human.

The audience know only one important information — the Robots tackles only the Humans, and the Humans tackles only the Robots.

Now, you are given a list of tackles, each involving two players. Based on this information, find the maximum possible number of Robots or Humans.

Input

The first line contains two integers N and M ($1 \leq N \leq 2 \times 10^5$, $1 \leq M \leq 3 \times 10^5$) — the number of players in the match and the number of tackles occurred during the match respectively.

The next M lines will contain two integers u_i, v_i ($1 \leq u_i, v_i \leq N$) — player u_i tackled player v_i . Each tackle between two players will be reported at most once.

Output

Print the maximum possible number of Robots or Humans.

Examples

Input

```
5 6
3 4
3 2
5 4
5 2
4 1
1 2
```

Output

```
3
```

Input

```
5 4
4 3
1 3
3 2
3 5
```

Output

```
4
```

Input

```
4 1
1 3
```

Output

3

Input

6 6
1 3
1 4
3 6
4 6
4 5
6 2

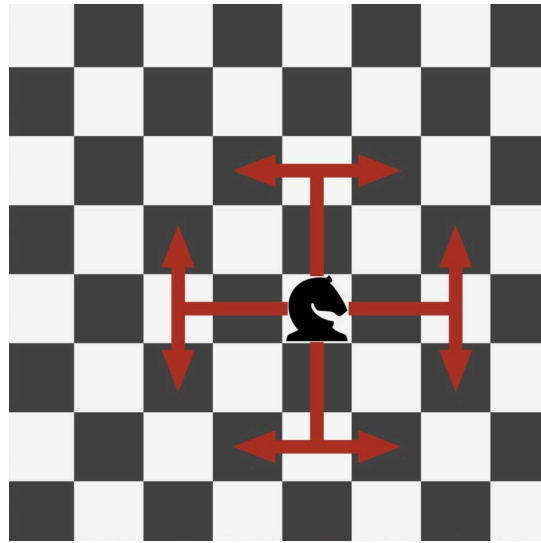
Output

3

C. The Knight of Königsberg

time limit per test: 15 seconds memory limit per test: 512 megabytes

You are given an $N \times N$ chessboard and the initial position (x_1, y_1) of a Knight piece. You need to find the minimum number of moves the Knight needs to reach the target position (x_2, y_2) . If it is not possible to reach the target, print -1 .



Moves of a Knight in Chess

The Knight can move one step in any of the 8 possible directions as shown in the picture.

Input

The first line contains an integer $(1 \leq N \leq 2 \times 10^3)$ — the size of the chessboard.

The second line contains four integers $(1 \leq x_1, y_1, x_2, y_2 \leq N)$ — the initial position (x_1, y_1) and the target position (x_2, y_2) of the Knight on the chessboard.

Output

Print the minimum number of moves the Knight needs to reach the target position. If it's not possible, print -1 .

Examples

Input

```
3
1 2 1 3
```

Output

```
3
```

Input

```
3
1 1 2 2
```

Output

```
-1
```

Input

```
10
8 4 3 1
```

Output

```
4
```

D. What's the Diameter?

time limit per test: 1 second memory limit per test: 1024 megabytes

You are given an undirected connected graph with N nodes and $N - 1$ edges. Your task is to find two nodes such that the path between those two nodes is the longest possible in the graph.

Input

The first line contains one integer N ($2 \leq N \leq 200000$) — the number of nodes.

The next $N - 1$ lines will contain two integers u_i, v_i ($1 \leq u_i, v_i \leq N$) — denoting there is a bidirectional road between u_i and v_i .

Output

On the first line, print a single integer — the length of the longest path.

On the second line, print two integers A and B — the nodes that form this longest path. If multiple pairs exist, you may print any one.

Examples

Input

```
5
5 1
1 4
4 2
3 2
```

Output

```
4
3 5
```

Input

```
5
1 2
5 3
3 2
2 4
```

Output

```
3
5 1
```

Input

```
8
1 7
7 3
3 6
6 5
5 2
2 8
8 4
```

Output

7
4 1

Input

7
7 5
5 6
6 1
1 3
3 4
4 2

Output

6
7 2

E. Nearest Tour Destination

time limit per test: 1 second memory limit per test: 256 megabytes

Forbidden Keywords for the Quiz: open, file, creat(, fstream, thread, process, system(, exec(

You are given an undirected unweighted graph with N nodes and M edges. The nodes are numbered from 1 to N . The graph contains no self-loops and no multiple edges.

There are S sources and Q destinations. For each destination node, find the length of the shortest path from any source node to that destination. If a destination is unreachable from all sources, output -1 .

Input

The first line contains four integers N, M, S, Q ($1 \leq N \leq 2 \times 10^5$, $0 \leq M \leq 3 \times 10^5$, $1 \leq S \leq N$, $1 \leq Q \leq N$) — the number of nodes, the number of edges, the number of source nodes, and the number of destination nodes.

The next M lines will contain two integers u_i, v_i ($1 \leq u_i, v_i \leq N$) — denoting there is an edge from node u_i to node v_i .

The next line contains S ($1 \leq S_i \leq N$) integers representing the source nodes, and the final line contains Q ($1 \leq Q_i \leq N$) integers representing the destination nodes. A node may appear both as a source and as a destination.

Output

The output should consist of Q integers separated by spaces. The j -th integer denotes the length of the shortest path from any source node to the j -th destination node. If no such path exists for a destination node, print -1 for that destination. A node may be both a source and a destination, in which case the answer for that destination is 0.

Examples

Input

```
8 6 2 4
1 2
2 3
4 5
6 7
7 8
2 6
1 6
3 5 6 8
```

Output

```
2 -1 0 2
```


Input

```
18 17 4 10
1 2
2 3
3 4
4 1
3 5
5 6
6 7
8 9
9 10
10 8
10 11
11 12
9 13
13 14
15 16
16 17
17 15
15 1 6 8
14 3 10 7 1 12 11 5 18 16
```

Output

```
3 2 1 1 0 3 2 1 -1 1
```

F. Digit Lock

time limit per test: 1 second memory limit per test: 256 megabytes

Alice has a treasure secured by a 4-digit numeric lock.



The 4-digit numeric lock.

Of course, Alice knows the correct combination C required to unlock this lock. However, there is an issue. Due to a mechanical fault in the numeric lock, certain combinations are deemed forbidden.

Alice wants to unlock this lock. In a single step, Alice can take one of the four "wheels" and rotate it by one step. More formally, Alice can take a single digit and either increment it or decrement it by 1 (modulo 10). The digits rotate around like clocks, so if we decrement 0 the digit becomes 9 and if we increment 9 the digit becomes 0. See the sample explanation for better understanding.

Alice wants to find the minimum number of steps required to reach the correct combination C from the starting combination S .

Input

The first line of the input contain the starting combination S and the correct combination C . Both are described in the input as 4-digit numbers. It is guaranteed that the starting combination is not forbidden.

The second line contains an integer n ($1 \leq n \leq 10^4$), the number of forbidden combinations.

The next n lines each contain a forbidden combination.

Output

If it is impossible to reach C , then output -1 . Otherwise, output the minimum number of required steps.

Examples

Input

```
0000 0003
1
0001
```

Output

```
5
```

Input

```
0000 0007
1
0006
```

Output

3

Input

1234 5678
12
2234
3234
4234
5234
1334
1434
1534
1634
1244
1254
1264
1274

Output

16

Note

In the first sample, our starting combination is 0000 and our target combination is 0003, while there is only one forbidden combination: 0001. So we can not directly increment the last digit in our starting combination. The following steps are optimal: 0000 → 0010 → 0011 → 0012 → 0013 → 0003.

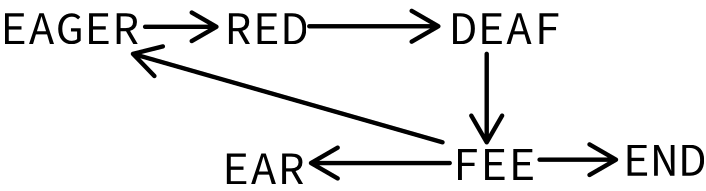
in the second sample, our starting combination is again 0000 while our target combination is 0007. The following steps are optimal: 0000 → 0009 → 0008 → 0007.

H. Word Chain

time limit per test: 1 second memory limit per test: 256 megabytes

Bob is playing the word chain game alone. In this game, he has a list of n ($1 \leq n \leq 10^5$) words $w_1, \dots, w_n$ and starts with word A (which is also in the list).

If Bob's current word is X , he can move to another word Y from the list if the last letter of X is equal to the first letter of Y . For example, if the current word is EAGER, then the next word can be RED (if it is in the list) as the last letter of EAGER and the first letter of RED are both R.



Graph for the first sample.

Bob is wondering if he can reach the word B (which is also in the list).

Input

The first line contains the integer n and the words A and B (starting word and target word). The i -th line of the next n lines contains w_i , the i -th word.

All words consist of uppercase Latin letters only and are non-empty. It is guaranteed that the sum of length of all the words in the list is at most 10^6 .

Output

In a single line, output YES if it is possible to reach word B from word A . Otherwise, output NO.

Examples

Input

```
6 EAGER EAR
EAGER
RED
FEE
DEAF
END
EAR
```

Output

```
YES
```

Input

```
5 EAGER EAR
EAGER
RED
DEAF
END
EAR
```

Output

```
NO
```

Note

For the first sample, Bob can reach EAR via the following path: EAGER → RED → DEAF → FEE → EAR.

In the second sample, the word FEE has been removed. Now, it is impossible to reach EAR from EAGER.

I. An Ancient Ordering

time limit per test: 1 second **memory limit per test:** 256 megabytes

You have found an old dictionary containing N words. The words are stored in an order that is different from the regular Latin lexicographic order.

Your task is to determine the order of the alphabet that satisfies the lexicographic order of this dictionary. If there are multiple valid orders, print the lexicographically smallest one. For example, the sequence $S_1 = \text{"d x i k"}$ is lexicographically smaller than the sequence $S_2 = \text{"d x p a k"}$.

If no such valid sequence exists, print -1 . A valid ordering is not possible if the characters create cyclic dependencies or if a longer word appears before a shorter word that is a prefix of it.

Input

The first line contains an integer N ($1 \leq N \leq 1000$) — the number of words in the dictionary.

The next N line contains a string S ($1 \leq |S| \leq 100$). Each word consists of only lowercase Latin letters $a - z$.

Output

Find out the order of the alphabets that satisfy the sorting order of the words in the given dictionary. If there are multiple valid orders, print the lexicographically smallest one. If no such valid sequence exists, print -1 .

Examples

Input

```
3
eat
tea
ate
```

Output

```
eta
```

Input

```
9
error
tooth
tot
teeth
their
there
thi
tie
hit
```

Output

```
oethir
```

Input

```
6
gef
gie
hf
hd
hc
ha
```

Output

```
efdcaghi
```

Input

```
5
cmwage
yent
jtdgx
wlp
xufjpf
```

Output

```
acdefglmnpqtuyjwx
```

Input

```
6
abc
ab
p
pq
pqr
pqrs
```

Output

```
-1
```

Input

```
2
pigeon
pigeons
```

Output

```
eginops
```

Input

```
4
ab
bc
ca
ac
```

Output

```
-1
```